

Roll No. :

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B.Tech**

Semester: V

Name of the Course : **Computer System Security**

Course Code: TCS 591

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1	(10 Marks)	CO1
(a)	How can a buffer overflow occur in a C program that lacks boundary checks on array operations? Suggest one sandboxing or isolation technique that can mitigate this attack	CO2
	OR	
(b)	In what ways can an integer overflow vulnerability be exploited to bypass browser memory protection and escalate privileges? Illustrate with a suitable scenario.	CO1
Q2	(10 Marks)	
(a)	What is a format string vulnerability? Give one example of how it can be misused.	CO2
	OR	
(b)	Construct an example of how a race condition vulnerability can be exploited in a file-handling service. How does the Dirty COW vulnerability in Linux demonstrate a similar issue?	CO1
Q3	(10 Marks)	
(a)	How can a format string vulnerability be escalated into code execution? What countermeasure at the compiler level could prevent such an exploitation?	CO2
	OR	
(b)	What are the key differences between static analysis and dynamic analysis in vulnerability detection? Which is more effective for runtime attacks?	CO1
Q4	(10 Marks)	
(a)	How does a Return-to-libc attack work? Suggest one simple defense mechanism against it.	CO2
	OR	
(b)	Compare Role-Based Access Control (RBAC) and Privilege Separation in operating system security. Which method is more effective against insider threats, and why?	CO1
Q5	(10 Marks)	
(a)	Design a secure web application model that integrates sandboxing, the CIA triad, and fuzzing for vulnerability testing. Present your answer with a labeled diagram and explanation.	CO2
	OR	
(b)	List three measures you would include in a secure web application model to ensure confidentiality, integrity, and availability.	

